

PGA-UNet: A Lightweight Prompt-Guided Attention U-Net for Interactive Bone X-Ray Lesion Segmentation

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ABSTRACT Bone X-ray lesion segmentation is a challenging task because lesions are often small, low-contrast, and visually entangled with the surrounding anatomy. In this setting, the main challenge typically lies in localizing the suspicious region rather than refining the lesion's boundary. A practical strategy is to narrow the search space before segmentation itself is performed. We realize this idea through a direct interactive approach: a clinician supplies a coarse bounding box, which the model then uses as a spatial guidance signal.

We propose PGA-UNet, a lightweight prompt-guided CNN that converts the bounding-box prompt into a Gaussian-smoothed plateau heatmap, modulates encoder features according to the prompt through a Prompt Spatial Gate (PSG), and sustains prompt influence in the decoder through a Conditional Attention Decoder (CAD). The goal is not only to use the prompt to indicate where segmentation should occur, but also to keep small-lesion-relevant features active throughout the network.

Attention U-Net serves as the automatic baseline for prompt-free segmentation, while SAM-Med2D provides a prompt-matched reference. The main experiments evaluate PGA-UNet on BTXRD and FracAtlas at multiple resolutions under controlled covering and off-center boxes. Small-lesion and ablation analyses examine whether the proposed prompt-conditioned design remains useful when the target area is limited. Because all reported boxes are simulated around annotated lesions, these experiments assess prompted segmentation after controlled localization rather than unconstrained lesion detection or clinical usability.

Component-contribution analysis evaluates PSG, CAD, and Gaussian prompt encoding within the tested ablation configurations. The efficiency analysis measures parameter count, FLOPs, memory, and latency in the specified benchmark environment. The resulting evidence is interpreted within the controlled prompted-segmentation protocol and does not establish calibrated confidence, clinical robustness, patient-level generalization, or cross-dataset transfer. The quantitative tables in the current manuscript draft are being refreshed after retraining under the standardized 80/20 `center_mixed` protocol and should be read accordingly.

INDEX TERMS bone X-ray, interactive segmentation, medical image segmentation, prompt-guided learning, Attention U-Net

I. INTRODUCTION

BONE X-ray imaging is widely used in screening, diagnostic support, and follow-up because of its speed, low cost, and broad accessibility. Lesion segmentation in radiographs, however, remains difficult when the lesion is very small, low-contrast, or partially obscured by surrounding anatomy. In these cases, one of the main challenges lies in localization: before any boundary can be refined, the model must first identify the suspicious region within the entire image.

This is why small-lesion segmentation is rarely just a boundary-delineation problem. When a lesion occupies only a tiny fraction of the image, searching the entire radiograph is both inefficient and error-prone. A natural response is to narrow the search space first, then segment within the reduced region. Fully automatic methods attempt to learn this process internally. Although U-Net and Attention U-Net [3], [4] remain representative models of this approach, they still require the network to localize and segment the lesion entirely on its own, without any external assistance.

An alternative is to draw region-of-interest guidance from an external source. In clinical practice, a clinician can easily draw a coarse bounding box around a suspicious area, something a fully automatic model can rarely detect reliably for very small lesions. In this sense, a bounding-box prompt is more than an interactive convenience; it is a practical means of injecting spatial knowledge into the pipeline before detailed segmentation begins. Prompt-based interactive segmentation methods have grown out of this idea, from clicks and scribbles [5], [6] to recent promptable foundation models such as SAM and SAM-Med2D [1], [2].

For the present task, however, one issue remains unresolved. Once a prompt has identified the region of interest, should it serve merely as a positional constraint for mask generation, or should it continue to shape feature extraction so that faint lesion signals survive downsampling? This distinction matters most for very small lesions, where a small spatial deviation can substantially affect the segmentation outcome.

We address this question with PGA-UNet, a lightweight prompt-guided CNN in which the box prompt is treated as a structured spatial prior rather than an auxiliary input channel. The prompt is encoded as a plateau-style Gaussian heatmap, which is then modulated into encoder features through a Prompt Spatial Gate (PSG) and reused in the decoder through a Conditional Attention Decoder (CAD). This design is intended to keep prompt-guided lesion features active throughout the network, rather than merely confining segmentation to a predefined box region.

This perspective also shapes how the experiments are interpreted. Because Attention U-Net receives no spatial guidance from a clinician, its role is only to show how difficult the problem becomes when localization must be fully automatic. SAM-Med2D, in contrast, is chosen as the direct comparison because it receives the same bounding-box prompt input. The central question here is therefore not whether prompts are useful in general, but whether a lightweight architecture can exploit a coarse clinician prompt more effectively by preserving weak lesion signals, rather than treating the prompt merely as a spatial constraint.

The main contributions of this article are as follows.

- We propose PGA-UNet, a lightweight prompt-guided architecture for interactive bone X-ray lesion segmentation.
- We design a prompt-conditioning mechanism that combines a plateau-style Gaussian prompt, an encoder-side Prompt Spatial Gate (PSG), and a decoder-side Conditional Attention Decoder (CAD).
- We formulate an offline training and online interactive inference framework, with explicit mathematical definitions for prompt encoding, encoder gating, and decoder conditioning.
- We evaluate the method on the BTXRD and FracAtlas datasets under covering and off-center prompt conditions, comparing against Attention U-Net and SAM-Med2D using Dice, a center-based localization score

(CBL), and HD95, together with Monte Carlo cross-validation and computational efficiency measurements.

- We report component-contribution analysis, efficiency evaluation, and small-lesion analysis that clarify where the proposed prompt-conditioning design provides the greatest benefit.

II. RELATED WORK

Small-lesion segmentation in medical imaging has long been constrained by a simple practical fact: before a model can segment a tiny target well, it must first narrow down where to look. In the fully automatic route, this search-space tightening is learned implicitly inside the network. U-Net established the canonical multi-scale encoder-decoder template for biomedical segmentation [3], and Attention U-Net later introduced attention gates on skip connections so that decoder-side fusion could suppress irrelevant spatial responses more selectively [4]. These models improved automatic segmentation quality, but they still assume that the network itself must localize and segment the lesion from the raw image alone.

Interactive medical segmentation developed in parallel from the observation that partial user guidance can simplify the problem substantially [5], [6]. Early methods used clicks, scribbles, and geodesic cues; later work incorporated boxes as practical coarse prompts. Conceptually, these methods move part of the localization burden from the model to the user. For tiny radiographic lesions, that shift is especially meaningful: a clinician may be able to indicate a suspicious region coarsely even when a fully automatic system would struggle to find it reliably in the whole image.

The next major step was promptable foundation models such as SAM, SAM-Med2D, and MedSAM, which generalized prompt-driven segmentation across many medical domains [1], [2], [9]. These models are flexible, but their prompt usage is architecturally different from lightweight task-specific CNNs, and their computational cost is much higher. More importantly for the present task, they still leave open a design question that becomes acute for very small radiographic lesions: once a prompt has already localized the approximate region, should the prompt merely indicate where the mask should be generated, or should it continue to shape feature extraction and decoding so that weak lesion evidence is not lost?

The present work is motivated by that gap. In grayscale bone radiographs, the problem is not only to know *where* the lesion roughly is, but also to preserve weak lesion evidence while suppressing surrounding clutter after the search space has already been tightened. A lightweight prompt-guided CNN may therefore benefit from making prompt information active from early encoding to late decoding instead of using the prompt only as an initial cue or positional restriction.

III. METHOD

A. OFFLINE AND ONLINE FRAMEWORK

The proposed system has two stages. In the offline stage, the model is trained on radiographs paired with lesion polygons.

TABLE 1. Representative prior work and the remaining gap addressed in this article.

Work	Principle	Method	Performance Evidence	Pros	Cons
U-Net [3]	Fully convolutional encoder-decoder	Multi-scale skip-connected segmentation network	Broad biomedical benchmarks; Dice-based overlap reporting	Simple, efficient, strong canonical baseline	No prompt guidance; all localization must be automatic
Attention U-Net [4]	Skip-feature filtering by attention gates	U-Net with gated skip connections	Medical image segmentation tasks with overlap metrics such as Dice and IoU	Stronger automatic baseline than plain U-Net	Attention is still learned without explicit prompt conditioning
DeepGeoS and related interactive methods [6]	User guidance reduces search space	Geodesic or interaction-aware segmentation	Interactive medical segmentation under user input	Better reflects clinical interaction	Interaction mechanism can be task-specific or expensive to maintain
SAM-Med2D / MedSAM [2], [9]	Promptable foundation segmentation	Large pretrained model with prompt-driven mask decoding	Multi-dataset medical evaluation with prompt-based metrics	Flexible and broadly applicable	Heavy; prompt influence is not designed as a lightweight end-to-end radiograph-specific prior
This work	Prompt as a dense spatial prior throughout the network	Gaussian plateau prompt + PSG + CAD in a lightweight CNN	BTXRD and FracAtlas; Dice, IoU, HD95, CBL, efficiency, small-lesion analysis	Lightweight, prompt-aware from encoder to decoder, evaluated under controlled prompt displacement	Still requires supervised masks and fixed-resolution preprocessing

Each polygon is converted into a training prompt by generating a simulated bounding box and then transforming that box into a Gaussian-smoothed plateau heatmap. The network learns to map the image-prompt pair to the binary lesion mask. In the online stage, a user provides a coarse bounding box around a suspected lesion on a new radiograph. The same prompt-construction procedure is applied, and the trained model predicts the final lesion mask.

The scientific contribution of this framework is that the prompt is not treated as an optional side input used only at the first layer. Instead, the prompt is encoded as a dense spatial prior and made active at both encoder and decoder stages. The intended benefit is improved behavior when the prompt is coarse, mildly off-center, or applied to a very small lesion.

B. PROBLEM SETTING

Given a grayscale radiograph $\mathbf{I} \in \mathbb{R}^{H \times W}$ and a user-provided bounding box $\mathbf{B}$, the goal is to predict a binary lesion mask $\hat{\mathbf{M}}$. Let $\mathbf{H}(\mathbf{B})$ be the prompt heatmap derived from the box. PGA-UNet predicts

$$\hat{\mathbf{M}} = \mathbf{1} [f_{\text{PGA}}(\mathbf{I}, \mathbf{H}(\mathbf{B}); \theta) \geq 0.5]. \quad (1)$$

C. PROMPT REPRESENTATION

The input prompt is not encoded as a hard binary rectangle. Instead, the box interior is rasterized into a plateau map at the original image resolution and its boundaries are softened with a fixed 31×31 Gaussian kernel, before the heatmap is resized and padded down to the target square resolution together with the image. The kernel size is not rescaled with the target resolution (256×256 or 512×512): applying it once at the original resolution, prior to downsampling, keeps the effective blur consistent relative to whichever square frame the network ultimately sees. The plateau covers exactly the (possibly expanded) box with no additional minimum margin; coverage of the region of interest is guaranteed by the box expansion described below, not by a separate margin step.

During evaluation, prompts are tested in two settings. A covering prompt scales the tight lesion box outward from its center by a fixed factor of 3.0. An off-center prompt first applies the same scaling and then adds a displacement

controlled by `shift_ratio=0.5`, while preserving lesion coverage. A single model is trained per dataset and resolution using `center_mixed`, which selects `center_shift` with probability 0.8 and `center_zoom` with probability 0.2. The two conditions are evaluated separately at test time. The task scope assumes that the user has identified the suspicious region and provides a box intended to cover that lesion; partial-coverage, negative, and unrelated-region boxes are outside this controlled prompted-segmentation protocol.

Let (x_1, y_1, x_2, y_2) denote the tight lesion box and let $w = x_2 - x_1$ and $h = y_2 - y_1$. The simulated covering box uses the fixed center scaling described above during training and evaluation. The off-center box uses the same scaled box and applies the deterministic test-time displacement rule implemented in `dataset.py`, while preserving lesion coverage. These values were selected through preliminary protocol experiments and then fixed before the reported evaluation; they are not presented as globally optimal hyperparameters. This simulation is not a substitute for radiologist-drawn prompts, but it provides a controlled way to measure sensitivity to the defined prompt displacement.

Algorithmically, prompt construction follows three steps: 1) derive a lesion box from the polygon annotation or user input and expand it as described above, 2) rasterize the expanded box into a plateau mask at the original image resolution, and 3) smooth the plateau boundary with the fixed Gaussian kernel before resizing to the target resolution. This sequence avoids the hard-edge dependence of a binary box while preserving a clear spatial prior.

D. PROMPT SPATIAL GATE

Let $\mathbf{x}^l$ denote the encoder feature map at level l . PSG modulates it with a prompt-derived attention map $\mathbf{A}_{\text{PSG}}^l$:

$$\tilde{\mathbf{x}}^l = \mathbf{x}^l \odot (1 + \alpha_{\text{PSG}}^l \mathbf{A}_{\text{PSG}}^l), \quad (2)$$

where $\odot$ is element-wise multiplication and α_{PSG}^l is learnable. In the implementation, α_{PSG}^l is initialized to 0.1 and clamped to $[0, 1]$. The residual form keeps the original feature stream available while emphasizing prompt-relevant regions.

E. CONDITIONAL ATTENTION DECODER

CAD extends decoder attention by conditioning the skip gate on prompt-encoded features. Let $\mathbf{g}^l$ be the decoder gating feature and $\mathbf{p}_{\text{enc}}^l$ the prompt encoding at the same scale. The conditioned gate is

$$\mathbf{g}'^l = \mathbf{g}^l + c^l \alpha_{\text{CAD}}^l w_l \mathbf{p}_{\text{enc}}^l, \quad (3)$$

where c^l is a decoder-derived prompt-use scalar, α_{CAD}^l is learnable, and w_l is a fixed depth coefficient. This design is intended to keep the decoder conditioned on the prompt when the box is not perfectly centered.

The prompt encoder in CAD uses two 3×3 convolutions with instance normalization and ReLU. The prompt-use scalar c^l is produced by global average pooling followed by a 1×1 convolution and a sigmoid nonlinearity. The decoder

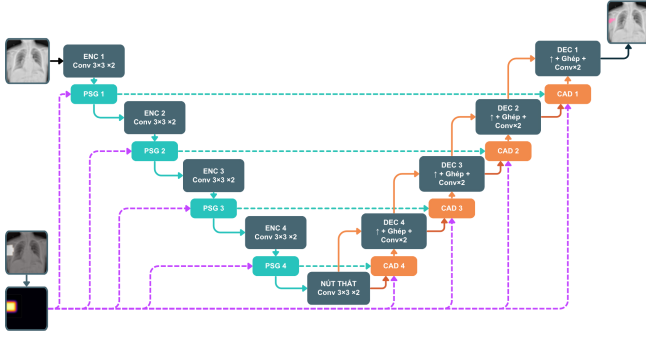


FIGURE 1. Overview of PGA-UNet. Prompt information derived from the bounding box is injected into encoder features through PSG and reused in decoder skip attention through CAD.

depth coefficients are fixed to $\{1.0, 0.7, 0.4, 0.2\}$ from deep to shallow layers. With feature scale 4, the channel widths are $\{16, 32, 64, 128, 256\}$ from the shallowest encoder stage to the bottleneck. These protocol values were selected in preliminary experiments and held fixed for the main evaluation; no optimality claim is made. The CAD mixing coefficient is parameterized as a sigmoid-transformed scalar and initialized near 0.3.

Taken together, the method can be summarized as follows: 1) preprocess the image and prompt into a common square resolution, 2) encode prompt-relevant spatial priors with PSG in the encoder, 3) propagate prompt-conditioned context through CAD in the decoder, and 4) predict the final binary lesion mask. The scientific meaning of the contribution is that prompt information is both spatially softened and repeatedly reused, rather than being injected once and left to dissipate.

IV. EXPERIMENTAL SETUP

A. DATASETS

We evaluate on two public bone radiograph datasets. BTXRD is a primary bone tumor X-ray dataset with polygon annotations for classification, localization, and segmentation [7]. Our BTXRD split contains 1,493/187/187 images with 1,848/238/232 lesion polygons for training, validation, and testing, respectively. FracAtlas is a fracture radiograph dataset with classification, localization, and segmentation labels [8]. After data cleaning and polygon normalization, our FracAtlas split contains 573/72/72 images with 730/95/92 lesion polygons for training, validation, and testing, respectively.

Splits are performed at the image level, and all polygons from the same image are kept in the same split. The public metadata were insufficient to verify strict patient-level separation. Each polygon is treated as one promptable lesion sample during training. For image-level evaluation, if an image contains multiple lesions, the model is run once per lesion box and the probability maps are merged by pixelwise maximum before thresholding at 0.5. Ground-truth masks are merged by union in the same image.

Inputs are resized isotropically and padded to square resolutions of 256×256 or 512×512 to preserve aspect ratio. Images, masks, and prompt maps undergo the same geometric preprocessing.

B. TRAINING DETAILS

All CNN models were implemented in PyTorch and trained on an NVIDIA Tesla T4 GPU with 16 GB memory. PGA-UNet and Attention U-Net were trained separately on each dataset and resolution. In other words, the article studies one shared architecture family instantiated as separate models with separate parameter sets for BTXRD and FracAtlas, rather than a single jointly trained cross-dataset model. The optimizer was AdamW with learning rate 10^{-4} and weight decay 10^{-4} . The loss was the sum of binary cross-entropy and Dice loss. Batch size was 4, the maximum training budget was 150 epochs, and early stopping with patience 15 was used. All main and ablation training runs use the fixed training seed 22120196. The primary model-selection criterion was image-level merged validation Dice under the `center_shift` condition.

Training-time augmentation included horizontal flipping and random rotation within $\pm 15^\circ$. The implementation also applied prompt dropout or prompt noise to a subset of training iterations to reduce over-reliance on an ideal prompt. PGA-UNet is trained with `center_mixed`, selecting `center_shift` with probability 0.8 and `center_zoom` with probability 0.2. The prompts are simulated from lesion annotations to provide a controlled prompted-segmentation protocol. The current article should therefore be read as a study under simulated lesion-covering boxes around user-identified lesions, not as an evaluation of unconstrained lesion detection or fully realistic radiologist-drawn boxes. For the fine-tuned SAM-Med2D comparison, the pretrained image encoder was frozen except for its lightweight Adapter layers, while the prompt encoder and mask decoder were fully updated on each dataset under the same split protocol. SAM-Med2D was fine-tuned with box prompts only, using the same center-scaled protocol as PGA-UNet; the point-prompt branch and iterative point refinement were disabled. Validation during SAM-Med2D fine-tuning and the final test split used both prompt scenarios, with batch size 4, a maximum of 150 epochs, early stopping patience 30, and learning rate 10^{-5} . The matched protocol reduces differences caused by prompt distribution and resolution, but does not eliminate differences in model scale, initialization, or optimization. Additional stability experiments were performed using Monte Carlo cross-validation, that is, repeated random image-level splits rather than strict non-overlapping k -fold partitioning. The Monte Carlo experiments keep the training seed fixed at 22120196 and vary only the split seed across 1, 2, 3, and 4.

Two evaluation details are important for interpreting the results. First, prompt generation itself is resolution-agnostic: the plateau heatmap is built and smoothed with a fixed 31×31 Gaussian kernel at the original image resolution, and only the resulting heatmap is resized and padded down to 256×256 .

or 512×512 together with the image, so the same absolute blur is compressed differently by the two target resolutions rather than being parameterized separately for each. Second, the Monte Carlo cross-validation experiments are auxiliary stability evidence rather than replacements for the fixed train/validation/test split used in the main tables.

C. BASELINES AND METRICS

PGA-UNet is compared with Attention U-Net [4] as the main automatic baseline and with SAM-Med2D [2] as the main prompt-based baseline. Attention U-Net is trained without prompts and is used to characterize how difficult the task is when localization must be solved automatically. SAM-Med2D is evaluated at 256×256 in fine-tuned form with the same dataset splits and the same box prompts as PGA-UNet, making it the primary prompt-matched comparison.

To further isolate whether PGA-UNet's advantage comes from its architecture or simply from having access to the box prompt at all, we add two prompt-matched conventional baselines built on the same Attention U-Net backbone, given the identical box prompt PGA-UNet receives but without PGA-UNet's Gaussian-prior heatmap, Prompt Spatial Gate, or Conditional Attention Decoder. Attention U-Net with a concatenated prompt channel appends the prompt heatmap as a second input channel and predicts on the full image through a plain skip-connection decoder. Attention U-Net on prompt crops instead restricts the network's field of view: the input image is cropped to the prompt box before resizing to the model's input resolution, the plain Attention U-Net predicts a mask on that crop alone, and the prediction is pasted back into the full frame for evaluation. Both baselines are trained under the same covering and off-center box protocol as PGA-UNet, so any remaining gap reflects the architecture rather than an unequal prompt.

The main paper emphasizes Dice, CBL, and HD95, while the code and exported reports also record IoU, precision, and recall. For a predicted mask centroid (x_p, y_p) , a ground-truth centroid (x_g, y_g) , and the diagonal length D_{GT} of the ground-truth bounding box, CBL is defined as

$$\text{CBL} = \max \left(0, 1 - \frac{\sqrt{(x_p - x_g)^2 + (y_p - y_g)^2}}{D_{GT}} \right). \quad (4)$$

HD95 is computed from bidirectional boundary distances. If both masks are empty, HD95 is set to 0. If exactly one mask is empty, HD95 is set to the image side length (256 or 512 pixels). No samples are discarded.

D. SCOPE OF ANALYSIS

This article focuses on prompt-guided segmentation itself and excludes the auxiliary screening pipeline explored separately outside the present manuscript. To keep the article compact, we retain only the subgroup analyses that most directly support the main claim: the role of prompt-guided localization, sensitivity to the controlled prompt displacement defined in this study, and effectiveness on very small lesions. Partial-coverage boxes, negative boxes, and unrelated-region boxes

are outside the present task scope and are therefore not used to support or reject the paper's claims.

V. RESULTS

Status note. The numerical tables currently shown in this section are legacy reference values from the previous protocol and are retained only to preserve the comparison structure while retraining is still pending. They have not yet been refreshed under the current standardized pipeline, which uses `center_mixed` training with 80% `center_shift` and 20% `center_zoom`, 150 training epochs, image-level merged validation selection under `center_shift`, and the current Precision definition. Until the retrained checkpoints and updated CSVs are in place, the text in this section should be read as a description of the intended comparison logic and of the claim scope, not as finalized evidence for the current manuscript version.

A. MAIN COMPARISON ACROSS DATASETS AND RESOLUTIONS

This first group of experiments addresses the most basic question: does PGA-UNet remain effective across both datasets, across both tested input resolutions, and across both prompt scenarios? In the refreshed manuscript, this subsection should validate the main system-level contribution, namely that prompt-conditioned processing through both encoder and decoder remains useful after the search space has already been tightened by a coarse box prompt. Because the current tables are legacy placeholders, this subsection should not yet claim reproduced performance for the standardized protocol. After retraining, the interpretation should remain limited to independent within-dataset evaluation on BTXRD and FracAtlas, not to cross-dataset generalization.

B. EFFECT OF INPUT RESOLUTION

This experiment isolates the contribution of image resolution within the same architecture and prompt protocol. It is meant to support the claim that the proposed method is especially useful when small-lesion evidence would otherwise be weakened after the search space has been reduced but the lesion signal itself is still fragile. PGA-UNet is trained independently at 256×256 and 512×512 on each dataset. Until retraining is complete, the specific numbers in Table 2 should not be cited as current-protocol evidence; only the intended comparison design should be carried forward.

The resolution comparison should not be overinterpreted as a pure image-size effect because the prompt heatmap is built once at the original image resolution with a fixed Gaussian kernel and only then downsampled together with the image, so the same absolute blur is compressed more aggressively into the 256×256 frame than into the 512×512 one. In the refreshed manuscript, the safest interpretation is that higher spatial detail may remain important for small-lesion prompt-guided segmentation under this protocol, especially when the lesion occupies only a tiny fraction of the image.

TABLE 2. PGA-UNet Dice under covering and off-center prompts at two resolutions.

Dataset	Resolution	Covering	Off-center
BTXRD	256 × 256	0.8449	0.8111
BTXRD	512 × 512	0.8788	0.8496
FracAtlas	256 × 256	0.7974	0.7619
FracAtlas	512 × 512	0.8286	0.7925

TABLE 3. Monte Carlo cross-validation for PGA-UNet at 512 × 512 (4 repeated random splits, mean ± standard deviation).

Dataset	Prompt	Dice	CBL	HD95
BTXRD	Covering	0.8715 ± 0.0083	0.9600 ± 0.0028	8.09 ± 0.67
	Off-center	0.8357 ± 0.0120	0.9390 ± 0.0062	10.45 ± 0.65
FracAtlas	Covering	0.8281 ± 0.0052	0.9597 ± 0.0039	6.93 ± 0.08
	Off-center	0.7862 ± 0.0114	0.9297 ± 0.0054	8.75 ± 0.48

C. MONTE CARLO CROSS-VALIDATION

This experiment examines whether the PGA-UNet results remain stable when the image-level split is redrawn repeatedly under limited data. It supports stability at the dataset-partition level rather than at the prompt-perturbation level. Supplementary experiments therefore used Monte Carlo cross-validation, that is, four repeated random image-level splits of the 512 × 512 PGA-UNet configuration, evaluated separately under covering and off-center prompts. These results are not intended to replace the fixed-split main tables; in the refreshed manuscript, this subsection should be limited to split-level stability of PGA-UNet under the evaluated protocol.

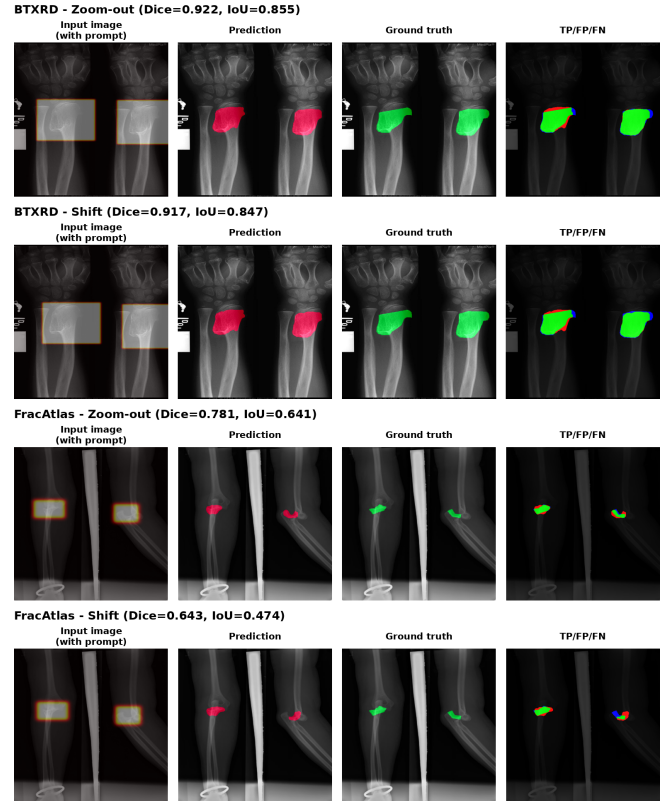
In the final refreshed version, this subsection should be used only to support split-level stability of PGA-UNet under the evaluated protocol. It should not be used by itself to claim superiority over any baseline. If paired statistical comparisons against baselines are later added, they should be presented separately rather than folded into this Monte Carlo section.

D. COMPARISON WITH THE AUTOMATIC BASELINE

Attention U-Net receives no clinician-provided box, so this comparison is not a like-for-like test of two prompt-guided methods. Its purpose is instead to measure how difficult this segmentation task is, over the full test set, when the network must localize the lesion entirely on its own, which is exactly the step that a coarse clinician prompt is meant to remove. Table 4 therefore serves as an automatic-reference comparison, not as a fully matched architecture comparison. In the refreshed manuscript, this subsection should support the narrower claim that automatic localization is a major challenge in this task setting, while avoiding any wording that would treat the Attention U-Net row as proof that PGA-UNet is architecturally superior in a like-for-like sense.

E. ATTENTION U-NET TOP- AND BOTTOM-DICE SUBSETS

This subgroup analysis is an extension of the automatic-baseline comparison. After comparing PGA-UNet with At-

**FIGURE 2.** Representative qualitative behavior under prompt perturbation. The figure illustrates the type of degradation observed when the prompt is shifted away from the ideal covering box, while also showing that thin fracture patterns remain harder than larger compact lesions.**TABLE 4.** Comparison with the automatic baseline at 512 × 512 under covering prompts.

Dataset	Model	Dice	CBL	HD95
BTXRD	Attention U-Net	0.5160	0.5995	129.67
	PGA-UNet	0.8788	0.9619	9.08
FracAtlas	Attention U-Net	0.3075	0.4658	110.83
	PGA-UNet	0.8286	0.9615	6.21

tention U-Net at the dataset level, we ask a more specific question: how does PGA-UNet behave on images where the automatic baseline is relatively strong versus relatively weak? The purpose is to clarify the practical meaning of the Attention U-Net comparison rather than to define objective data difficulty. We therefore compare Attention U-Net top-Dice and bottom-Dice image groups on both BTXRD and FracAtlas. Specifically, for each dataset we select the 50 test images with the highest Attention U-Net Dice scores and the 50 test images with the lowest Attention U-Net Dice scores, then re-evaluate PGA-UNet on the same two groups. This analysis does not define intrinsic data difficulty; it only reports performance on subsets induced by the Attention U-Net ranking.

Because the four groups are defined directly from Atten-

TABLE 5. Top-Dice and bottom-Dice comparison under covering prompts at 512×512 (50 images per group).

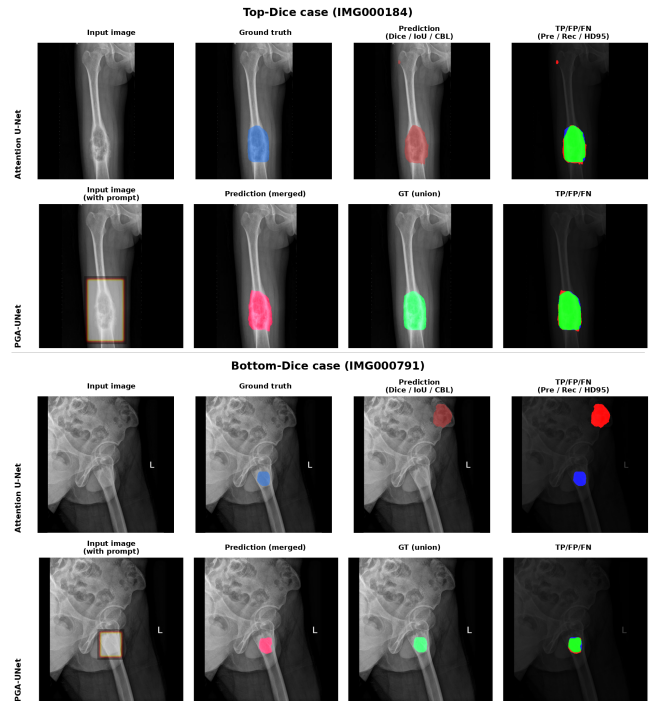
Dataset	Group	Model	Dice	CBL	HD95
BTXRD	Top-Dice	Attention U-Net	0.8629	0.9436	39.32
		PGA-UNet	0.9036	0.9709	9.23
	Bottom-Dice	Attention U-Net	0.0335	0.0928	270.72
		PGA-UNet	0.8521	0.9459	5.19
FracAtlas	Top-Dice	Attention U-Net	0.4428	0.6277	82.08
		PGA-UNet	0.8318	0.9666	6.37
	Bottom-Dice	Attention U-Net	0.1529	0.3345	135.76
		PGA-UNet	0.8151	0.9582	6.25

tion U-Net scores, these values should not be interpreted as global averages or as intrinsic clinical difficulty strata. In the refreshed manuscript, this subsection should keep that narrower interpretation explicit: the top-Dice and bottom-Dice groups only show how the prompt-guided models behave on image stems where the automatic baseline is relatively stronger or weaker.

The qualitative examples help clarify this behavior. In the Attention U-Net top-Dice groups, the automatic baseline can often place the predicted mask in roughly the correct anatomical region, but it may still produce spurious responses far from the lesion. Even when the lesion itself is covered reasonably well, those distant false positives reduce overlap-based metrics such as Dice and IoU, which is consistent with the still-elevated HD95 values (39.32 on BTXRD, 82.08 on FracAtlas) despite a moderate-to-low Dice. PGA-UNet is less prone to this pattern because the bounding box prompt constrains the model to a more relevant spatial region. In the bottom-Dice groups, the failure is more severe: the automatic baseline may localize the wrong region or miss the lesion almost entirely. Within the scope of the present evidence, the safest interpretation is not a definitive account of the Attention U-Net internals, but a system-level one: once the model must solve localization without an external prompt, confusing background structure and weak lesion evidence can lead to qualitatively different failure patterns from those seen in the prompt-guided setting. In contrast, PGA-UNet receives a coarse spatial prior and can therefore identify the lesion region more reliably before refining the mask.

F. MATCHED COMPARISON WITH SAM-MED2D

This comparison tests the architectural contribution of the proposed prompt handling against a prompt-based foundation baseline under matched input conditions. Both models receive the same prompt type and the same resolution, so the difference is more directly tied to how prompt information is represented and propagated after localization has already been externally provided. Table 6 reports the intended matched comparison at 256×256 under both covering and off-center prompts. Because the present table still contains legacy values, this subsection should currently be interpreted as the comparison structure and claim scope only; the final wording about relative performance against SAM-Med2D must be decided after retraining.

**FIGURE 3.** Qualitative comparison between Attention U-Net and PGA-UNet on BTXRD for one case from the Attention U-Net top-Dice group (IMG000184) and one case from the Attention U-Net bottom-Dice group (IMG000791). Each row shows four panels; for PGA-UNet, the input-image and box-prompt panels are merged into a single "Input image (with prompt)" panel.**TABLE 6.** Dice comparison with fine-tuned SAM-Med2D at 256×256 .

Dataset	Prompt	PGA-UNet	SAM-Med2D-FT
BTXRD	Covering	0.8449	0.7624
	Off-center	0.8111	0.7542
FracAtlas	Covering	0.7974	0.7439
	Off-center	0.7619	0.7275

When the tables are refreshed, the relative degradation from covering to off-center prompts should be reported separately from absolute accuracy. The two prompt conditions measure sensitivity to the controlled displacement defined in this study; they do not establish robustness to arbitrary clinician prompts. Any comparison with SAM-Med2D should therefore report both absolute Dice and the change between the two conditions without assuming that the model with the higher absolute Dice also has the smaller relative degradation.

G. PERFORMANCE ON SMALL LESIONS

This experiment is an extension of the SAM-Med2D comparison to the setting in which the contribution of prompt-aware feature preservation should be most visible, namely very small lesions. To keep subgroup analysis objective, we retain only the small-lesion subset because it is defined directly by lesion size rather than by a model-dependent ranking rule. For each dataset, the subset is formed by selecting the 50 test

TABLE 7. Dice, CBL, and HD95 on the 50 test images with the smallest total lesion area per dataset. For the R256 comparison, the reported metrics are computed in a shared 512×512 reference frame. IoU is omitted here because it is a noisy, boundary-sensitive statistic at this lesion size; Dice, CBL, and HD95 provide complementary overlap, centroid, and boundary measures.

Dataset	Model	Dice	CBL	HD95
BTXRD	SAM-Med2D-FT 256×256	0.1903	0.4489	240.72
	PGA-UNet 256×256	0.7650	0.9261	3.11
	PGA-UNet 512×512	0.8483	0.9451	2.32
FracAtlas	SAM-Med2D-FT 256×256	0.3736	0.7116	105.19
	PGA-UNet 256×256	0.7973	0.9569	5.62
	PGA-UNet 512×512	0.8348	0.9609	4.83

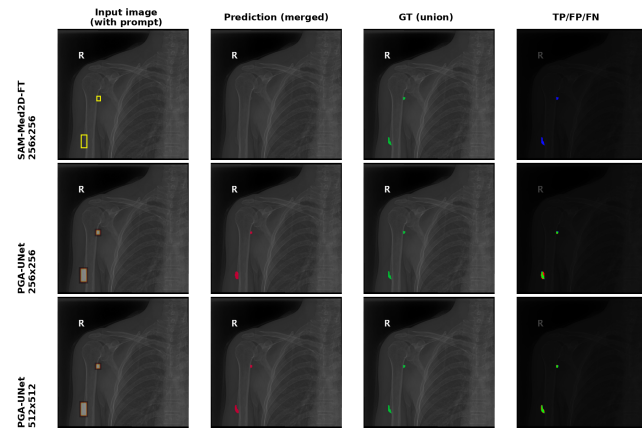


FIGURE 4. Representative qualitative comparison on small lesions in BTXRD (IMG000868, two small polygons). The examples contrast fine-tuned SAM-Med2D at 256×256 , PGA-UNet at 256×256 , and PGA-UNet at 512×512 ; the input-image and box-prompt panels are merged into a single "Input image (with prompt)" panel. The figure illustrates the type of small-lesion behavior captured by the quantitative comparison in the shared evaluation frame.

images with the smallest total lesion area, evaluated under covering prompts. This subset is then used for two related comparisons: PGA-UNet versus fine-tuned SAM-Med2D at the matched resolution of 256×256 , and PGA-UNet₂₅₆ versus PGA-UNet₅₁₂ to examine the effect of resolution. All metrics are computed at a shared 512×512 reference frame, with the 256×256 predictions of SAM-Med2D and PGA-UNet upsampled before scoring, so that the three models are compared on identical ground-truth masks.

In the refreshed manuscript, this subsection should remain focused on the narrow small-lesion question and should avoid overextending from subgroup results to broad clinical robustness. The safest interpretation is that very small lesions are an especially relevant stress case for prompt-conditioned segmentation, and that both explicit prompt conditioning and higher spatial resolution may matter more strongly there than on the full test set.

H. EFFICIENCY ANALYSIS

This experiment evaluates whether the proposed gains are achieved with a practical computational profile. Table 8 reports the efficiency comparison, measured on the same

TABLE 8. Efficiency comparison in the tested environment.

Model	Params	GFLOPs	Size	GPU ms	CPU ms
PGA-UNet 256	2.95M	7.74	11.35 MB	8.17	109.17
PGA-UNet 512	2.95M	30.97	11.35 MB	18.28	336.74
SAM-Med2D 256	271.24M	92.02	2443.16 MB	48.75	1263.77

TABLE 9. Ablation results (Dice) on BTXRD at 512×512 .

PSG	Decoder attention	Prompt	Covering	Off-center
No	None	Gaussian	0.8596	0.8230
No	None	Binary	0.8597	0.8253
No	Vanilla gate	Gaussian	0.8671	0.8408
No	CAD	Gaussian	0.8685	0.8358
Yes	None	Gaussian	0.8567	0.8297
Yes	Vanilla gate	Gaussian	0.8611	0.8314
Yes	CAD	Binary	0.8611	0.8317
Yes	CAD	Gaussian (PGA-UNet)	0.8788	0.8496

NVIDIA Tesla T4 GPU used for training (mean latency over repeated forward passes, batch size 1). PGA-UNet contains only 2.95M parameters. In the tested 256×256 setting, this corresponds to an approximately 92-fold reduction in parameter count and an 11.9-fold reduction in FLOPs relative to SAM-Med2D, together with $6.0\times$ faster GPU inference and $11.6\times$ faster CPU inference. The safe claim here is limited to lower measured computational cost and latency in the tested implementation, hardware, and input-resolution setting.

I. ABLATION STUDY

The ablation study directly tests which proposed components are responsible for the observed gains. In particular, it examines whether performance comes from the prompt representation itself, from encoder-side prompt gating (PSG), from decoder-side prompt-conditioned attention (CAD), or from their combination. We isolate three factors: whether PSG is present in the encoder, what form of decoder skip-attention is used (none, the original unconditioned Attention U-Net gate, or CAD), and whether the prompt is represented as a binary box or a Gaussian-smoothed plateau. Combining these factors gives eight trained configurations per dataset, all at 512×512 : the seven partial variants that isolate individual factors and their pairwise interactions, plus the full proposed PGA-UNet, which uses PSG, CAD, and the Gaussian prompt together. Tables 9 and 10 report Dice under covering and off-center prompts for BTXRD and FracAtlas, respectively.

The full PGA-UNet configuration (PSG, CAD, and the

TABLE 10. Ablation results (Dice) on FracAtlas at 512×512 .

PSG	Decoder attention	Prompt	Covering	Off-center
No	None	Gaussian	0.7727	0.7356
No	None	Binary	0.7670	0.7199
No	Vanilla gate	Gaussian	0.7560	0.7311
No	CAD	Gaussian	0.8158	0.7777
Yes	None	Gaussian	0.7115	0.6990
Yes	Vanilla gate	Gaussian	0.7372	0.7170
Yes	CAD	Binary	0.8132	0.7896
Yes	CAD	Gaussian (PGA-UNet)	0.8286	0.7925

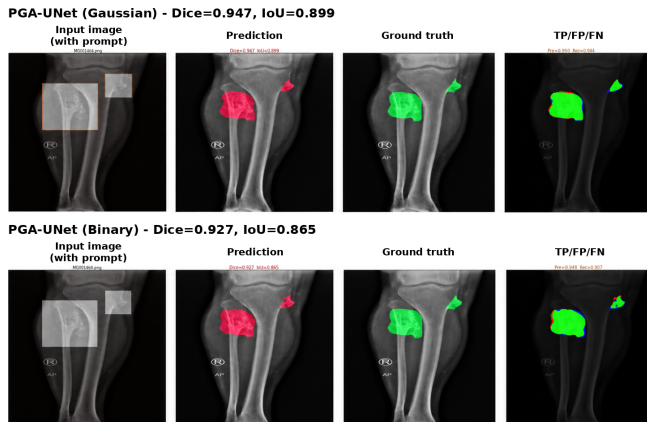


FIGURE 5. Representative qualitative comparison between the Gaussian prompt representation and the binary prompt representation. The figure illustrates a case in which the Gaussian version yields cleaner boundaries and better overlap.

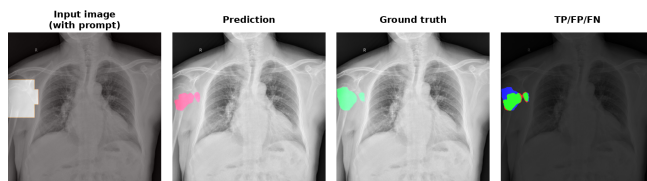


FIGURE 6. Representative failure case in an anatomically overlapping region. Even with a covering prompt, the lesion boundary remains difficult because the local X-ray structures are visually confounded.

Gaussian prompt combined) is the intended reference row in this ablation design. Once retrained under the current protocol, the safest wording is that the resulting pattern may be consistent with complementary contributions from PSG, CAD, and Gaussian prompt encoding within the evaluated configurations. Without repeated multi-seed ablation runs or factorial statistical analysis, the manuscript should avoid stronger language such as “proves synergy” or any claim of definitive mechanistic decomposition.

J. FAILURE MODES

The method still has identifiable failure cases. In the qualitative review collected from the repository’s failure-case examples, lower Dice scores were more likely for lesions with highly elongated or branching shapes, lesions located in anatomically cluttered regions such as the shoulder or clavicle, and lesions with weak contrast relative to surrounding structures. These cases are relevant because they show that prompt guidance does not eliminate ambiguity once the local image evidence itself is poor.

VI. DISCUSSION

The results are intended to support a fairly coherent interpretation of the task once the current-protocol tables are refreshed. First, the main question is not whether prompting is useful in the abstract, but whether prompt-conditioned processing remains useful after a coarse lesion-covering box has already narrowed the search space. Second, Attention

U-Net receives no clinician-provided box, so its result on both datasets is not a competing prompt-guided method; it instead shows how much of the difficulty in this task comes from localization alone. In that sense, the Attention U-Net comparison should be read as evidence that automatic localization is a major challenge in this setting, not simply as a ranking of network architectures. Third, the prompt-matched comparison that actually tests the proposed design is against fine-tuned SAM-Med2D at the matched 256×256 resolution, with both models evaluated under the same controlled `center_zoom` and `center_shift` conditions and trained under the same 80/20 `center_mixed` prompt distribution. Fourth, the small-lesion subset is retained because it is the subgroup where prompt-aware feature preservation is most plausibly relevant and because it is defined by lesion area rather than by a model-dependent ranking rule. Fifth, the ablation study is intended to test whether the evaluated combination of PSG, CAD, and Gaussian prompt encoding is useful as a system design, while the efficiency analysis addresses whether that design maintains a lower measured computational cost under the tested implementation, hardware, and input-resolution setting.

The scope of the analysis was deliberately narrowed. The auxiliary screening pipeline is excluded because it addresses a different question from prompt-guided segmentation itself. Likewise, the retained subgroup analyses are objective constructions tied either to lesion size or to a baseline-defined ranking.

The current study still has further limitations. The model depends on resizing inputs to a fixed resolution, which can dilute fine lesion detail in very small targets. Prompts are simulated from existing annotations rather than collected from radiologists, and the defined protocol assumes that the user has already identified the suspicious region and provides a box intended to cover it. Partial-coverage boxes, negative boxes, and unrelated-region boxes are outside the present task scope by design, so the article should not imply that those settings were tested. The current formulation does not yet exploit additional information sources beyond the image and box prompt, and the training setup still requires dense mask supervision. The evaluation also remains limited to the current public datasets, with BTXRD and FracAtlas trained separately rather than jointly or evaluated through external-domain transfer. Patient-level separation could not be verified from the available public metadata. The Monte Carlo cross-validation experiments cover split-level stability of PGA-UNet, but they do not by themselves establish superiority over any baseline. The ablation study is still reported from a single deterministic split rather than from repeated multi-seed training.

Another limitation concerns the optimization and evaluation components. The current loss does not yet emphasize the small-lesion or accurate-center-localization properties that this task actually needs, even though CBL is already used at evaluation time. The prompt scale, displacement ratio, Gaussian kernel, and CAD depth weights are fixed protocol

values selected through preliminary experiments; they are not claimed to be globally optimal. The present article therefore demonstrates the system-level behavior of PGA-UNet under the selected protocol, but has not yet explored the effect of alternative prompt or loss configurations systematically.

An additional limitation is interpretive rather than purely empirical. Several observed gains, especially under prompt perturbation, are consistent with the intended roles of PSG, CAD, and the Gaussian plateau prompt, but this article does not include feature-level probing that would establish a unique internal mechanism for those gains. The ablation study therefore supports the usefulness of the design choices at the system level, not a definitive causal account of representation dynamics inside the network.

VII. CONCLUSION

This article presented PGA-UNet, a lightweight prompt-guided CNN for interactive bone X-ray lesion segmentation. The central design treats a user-supplied lesion-covering box as a dense spatial prior and reuses it through encoder-side PSG and decoder-side CAD. The manuscript is scoped as a controlled prompted-segmentation study: it compares automatic and prompt-matched baselines, evaluates controlled covering and off-center prompts, and examines small-lesion behavior, repeated image-level splits, ablations, and computational cost within that defined protocol. The conclusions are therefore limited to simulated lesion-covering boxes derived from annotations and do not establish unconstrained detection, performance for partial or negative boxes, radiologist-provided prompt behavior, calibrated confidence, patient-level generalization, or cross-dataset transfer. Future work should refresh all current-protocol quantitative tables after retraining, evaluate broader prompt conditions if the task scope is expanded, and validate the auxiliary QualityHead score before interpreting it as a calibrated probability.

DATA AVAILABILITY

BTXRD and FracAtlas are public datasets cited in this article. The manuscript reports experiments on image-level train/validation/test splits derived from those datasets. Code and split details should be released with the final reproducibility package.

AUTHOR CONTRIBUTIONS

Thong Luc: conceptualization, methodology, software, investigation, formal analysis, visualization, writing of the original draft. Nguyen Huu Binh: software support, data preparation, experiment verification, review and editing. Ly Quoc Ngoc: supervision, validation, review and editing.

ETHICS STATEMENT

This study used public de-identified radiograph datasets and did not involve prospective data collection, patient contact, or intervention.

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OpenAI ChatGPT was used only for vocabulary and grammar refinement in selected non-technical manuscript text. All technical content, experimental design, mathematical formulation, figures, tables, and conclusions were prepared and verified by the authors, who take full responsibility for the final manuscript.

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